A moment obstruction and minimal-degree polynomial nonconforming Stokes elements on tetrahedra

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Abstract

Nonconforming finite elements for incompressible flow are often specified by *moments*: averages of the velocity against polynomials on each face and inside each cell. A recent paper by Balazi, Allaire, Jolivet and Omnes builds tetrahedral Stokes elements of second and third order in this way and leaves open whether the fourth-order element can be obtained by adding fifteen quintic polynomials to the quartics. We show that it cannot, and we explain why. For moments of degree n on faces and $n-1$ in the cell, we compute exactly which polynomials of degree $n+2$ are invisible to all moments: for every n they form a seven-dimensional space with an explicit basis. Consequently degree $n+2$ suffices only for $n = 1, 2$; for $n \geq 3$ exactly $(n-1)(n-2)/2$ moment directions are missing, and degree $n+3$ is necessary and sufficient. These degree statements concern single-cell polynomial spaces that contain all polynomials of degree $n+1$ and are unisolvent for exactly these moments. At fourth order the single missing direction is detected by a short symmetry argument and supplied by the Vandermonde polynomial of the barycentric coordinates; the result is a 50-dimensional element that does not depend on vertex numbering. For every n we prove that numbering-free elements using the smallest possible number of top-degree functions exist; we give them explicitly for $n = 4, 5$ and check an explicit family up to $n = 8$. In two dimensions the analogous count shows that no extra degree is ever needed, which explains why triangular constructions extend to all orders. A numerical Stokes test shows rates consistent with fourth-order convergence; the computed discrete inf-sup values are 0.40–0.37 for the four coarsest meshes.

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1 Introduction

1.1 What moment degrees of freedom are for

A finite element is a local polynomial space together with a list of linear functionals, the *degrees of freedom*, that determine each local function uniquely. When neighbouring cells share the degrees of freedom attached to a common face, the pieces are glued together into a global space. If the shared quantities are point values at vertices, the global functions are continuous. If they are *moments*, that is, integrals $\int_F v\, q\, ds$ against test polynomials q on a face F, the global functions may jump across faces but the jumps are orthogonal to the tests. Such elements are called *nonconforming*; the lowest-order example is the Crouzeix–Raviart element, whose degrees of freedom are face averages [11].

Nonconforming elements are attractive for the Stokes equations of slow viscous flow, where a velocity $\mathbf{u}$ and a pressure p are sought together. A velocity–pressure pair is usable only if it is *inf-sup stable*: every discrete pressure must be detected by the divergence of some discrete

velocity, with a constant independent of the mesh [6]. Face and cell moments make this almost automatic. Integration by parts writes $\int_K q \operatorname{div} \mathbf{v}$ in terms of face moments of $\mathbf{v}$ against q and cell moments against ∇q. If the element's degrees of freedom include these moments, the interpolant that matches them preserves the divergence pairing, and stability follows from the continuous problem. The local algebra must then deliver a polynomial space on which the chosen moments are *unisolvent* (they determine the function uniquely) and which contains enough polynomials for the desired accuracy.

1.2 The question

Balazi, Allaire, Jolivet and Omnes [3] use, for an integer $n \geq 1$, the moments against $\mathbb{P}_n$ on each face of a tetrahedron and against $\mathbb{P}_{n-1}$ inside it, together with a discontinuous $\mathbb{P}_n$ pressure. To reach order $n+1$ the velocity space must contain $\mathbb{P}_{n+1}$. Counting shows that $n(n+2)$ additional functions are needed, and the source searches for them among polynomials of degree $n+2$. Its Remark 3.2 observes that this is dimensionally possible only for small n: "the proposed approach would work in theory at most for $n \leq 5$." The source constructs elements for $n = 1, 2$ by searches over monomials and notes in Remark 3.6 that "none of these bases are symmetric, that is to say, the finite element space depends on the face numbering." For $n = 3$, Appendix A of the open version [3] reports:

> "We test randomly millions of possible space $V_4(K)$, but as of today, no suitable space $\Sigma_5(K)$ has been found. The existence (or not) of such bases is still an open question."

(In the source's notation $V_4(K)$ is the fourth-order velocity space and $\Sigma_5(K)$ is the quintic enrichment.) The question is whether *some* 15-dimensional space of quintics, not only a set of monomials, enriches $\mathbb{P}_4$ to a unisolvent space.

1.3 The answer in words

No such space exists. The reason is not the choice of quintics but the quintics themselves: on all of $\mathbb{P}_5$ the 50 moments satisfy one linear relation, so they can never be independent there. A single polynomial of degree six, the product of the six differences $\lambda_j - \lambda_i$ of barycentric coordinates, breaks the relation.

More generally, for every order n we find all polynomials of degree $n+2$ whose moments vanish. There are always exactly seven independent ones, each a classical orthogonal polynomial in a single barycentric combination (a "ridge"). A dimension count then shows that degree $n+2$ leaves $(n-1)(n-2)/2$ moment directions unreachable; this is zero precisely for $n = 1, 2$. Degree $n+3$ always suffices. Remark 3.2 of [3] therefore sharpens from "at most $n \leq 5$" to "exactly $n \leq 2$". In two dimensions the same count gives zero missing directions for every n.

1.4 Contributions

Each result is labelled by the kind of evidence behind it: [proof], [exact computation] (rational or modular arithmetic, or exact symbolic identities, with a replayable script), [numerical] (floating point), or [conjecture].

1. [proof] The kernel of the face-and-cell moment map on $\mathbb{P}_{n+2}$ is seven-dimensional for every $n \geq 1$, with an explicit ridge basis (Theorem 2.1). The missing-direction count is $(n-1)(n-2)/2$; the moments are surjective on $\mathbb{P}_{n+3}$; the minimum degree is $n+2$ for $n = 1, 2$ and $n+3$ for $n \geq 3$.
2. [proof] A one-line symmetry argument explains the fourth-order failure and identifies the repairing sextic Δ_4 (Proposition 3.2).

3. [proof] At least $(n-1)(n-2)/2$ top-degree directions are needed, and vertex-numbering-invariant elements attaining this number exist for every n (Corollary 5.2, Theorem 5.4). The missing directions form the representation $\mathrm{sgn} \otimes \mathrm{Sym}^{n-3}(\mathrm{std})$ of the symmetric group [exact computation of a generating-function identity]. An explicit lean element for $n = 4$ uses three degree-7 functions $\lambda_i \Delta_4$ [exact computation]; the analogous choice $\Delta_4 \mathbb{P}_{n-3}$ works for $n \leq 8$ [exact computation] and is conjectured for all n [conjecture].

4. [proof] An explicit 50-dimensional fourth-order element with a 7×7 projection formula and a short proof of a 14-monomial basis (Section 5.5).

5. [proof] On a d-simplex the kernel has dimension $\sum_{r=1}^{\min(d,n+2)} \binom{d}{r}$, the missing-direction count has a closed form that vanishes exactly when $d = 2$ or $n \leq 2$, and the minimum degree lies between $n+3$ and $n+d$ for $n \geq 3$, $d \geq 3$. [exact computation] The minimum is $n+3$ for $d = 4$ ($n \leq 5$), $d = 5$ ($n \leq 4$), $d = 6$ ($n = 3$); [conjecture] it is $n+3$ for all $d \geq 3$.

6. [numerical] A manufactured-solution Stokes test with the fourth-order element shows rates close to 4 for the velocity gradient and the pressure, and computed discrete inf-sup values 0.40–0.37 on four successive meshes.

The Fortin and error estimates of Section 7 apply the established framework of [11, 3] and are included to fix hypotheses.

1.5 Relation to the literature

Two dimensions. Fortin and Soulié [14] introduced a quadratic nonconforming triangle for Stokes flow. Matthies and Tobiska [18] constructed inf-sup stable nonconforming triangular elements of every order by enriching complete polynomial spaces, and Baran and Stoyan [4] gave a stable higher-order family based on Gauss–Legendre continuity. The source [3] follows this template in three dimensions. Corollary 6.3 explains why the triangle never needs an extra degree, whereas the tetrahedron does from fourth order on.

Three dimensions. Orthogonal polynomials and vertex symmetry in tetrahedral Crouzeix–Raviart spaces are developed by Ciarlet, Dunkl and Sauter [10] and by Sauter and Torres [19]. Chen, Hu and Zhang [9] analyse low-order tetrahedral face-moment kernels; their cubic and quartic vertex polynomials are proportional to our vertex ridges. Ye and Zhang [23] prove injectivity of the same face and cell data on $\mathbb{P}_{n+1}$ (their $k = n-1$) and build a lifting that first projects data onto the image of $\mathbb{P}_{n+1}$; a finite element requires surjectivity onto all data, which is what we determine. Bohne, Ciarlet and Sauter [7, §6.3, Thm. 35] prove an obstruction to splitting Crouzeix–Raviart degrees of freedom into facet and simplex integrals for odd degree at least three in dimension at least three, with facet tests one degree below the shape functions; our moment pattern and question (the rank of the combined map and the resulting ambient degree) are different, but both results show that three dimensions are more rigid than two.

Other routes to stability. Conforming elements enrich Lagrange spaces by face bubbles (Bernardi and Raugel [5]) or use split meshes and exactly divergence-free spaces (Scott and Vogelius [20]; Guzmán and Neilan [15, 16]). Nonconforming virtual elements [8] reach every order with moment degrees of freedom by allowing nonpolynomial local functions. Our degree bounds say nothing about such split-cell or nonpolynomial spaces. Finite element exterior calculus [2] treats degrees of freedom and unisolvence for polynomial differential forms systematically; its degrees of freedom for scalar functions are vertex, edge, face and cell moments, not the face-and-cell pattern studied here. The simplex Jacobi operator used below is the zero-parameter case of the operator in Aktaş and Xu [1].

1.6 Roadmap and notation

Section 2 states the main theorem with a worked example. Section 3 gives the short symmetry proof of the fourth-order obstruction. Section 4 computes the kernel. Section 5 proves the sharp degree and builds symmetric elements, including the lean ones and the explicit fourth-order element. Section 6 treats triangles and d-simplices. Sections 7–9 record the Stokes consequences, the numerical test and implementation advice. Table 1 collects the notation.

Table 1: Notation. Symbols are used only in the meanings listed.

K, F_i, λ_i	tetrahedron, face $\{\lambda_i = 0\}$, barycentric coordinates $\lambda_0, \dots, \lambda_3$
$\mathbb{P}_r(S)$	polynomials of total degree $\le r$ on a set S
$\fint_S$	normalised integral $\lvert S\rvert^{-1} \int_S$
n, $m = n+2$	moment order; degree of the kernel polynomials
$\mathcal{M}_n$, N_n	moment map (face tests $\mathbb{P}_n$, cell tests $\mathbb{P}_{n-1}$) and number of moments
c_n	number of missing moment directions on $\mathbb{P}_{n+2}$
Z_m	kernel of $\mathcal{M}_n$ on $\mathbb{P}_m$
$J_r^{(\alpha,\beta)}$, J_r	Jacobi polynomials; Legendre $J_r = J_r^{(0,0)}$
U_i, V_{0j}	vertex and pair ridges (kernel basis)
Δ_4, $\Delta_3^{(i)}$, ε_i	Vandermonde of $\lambda_0, \dots, \lambda_3$; of the three coordinates other than λ_i; signed cofactor (8)
$\mathcal{A}$	alternating face functional
$\mathcal{E}, \mathcal{D}, \mathcal{L}$	Euler, degree-lowering and simplex Jacobi operators
H, H'	homogeneous parts of degrees m, $m-1$
∂_{ij}, a_B	tangent derivative $\partial/\partial\lambda_i - \partial/\partial\lambda_j$; invariant energy form
Γ	Gram matrix of the ridges under a_B
V_n, W	local shape space; ambient space
$\boldsymbol{\eta}_F$, ν	unit outward normal to F; viscosity
d, $N_{d,n}$, $k(d,m)$, $c(d,n)$	simplex dimension, moment count, kernel dimension, missing directions

2 Setting and main theorem

Fix $n \ge 1$. The moments are

$$\fint_{F_i} v\, q\, ds, \quad q \in \mathbb{P}_n(F_i),\ i = 0,1,2,3; \qquad \fint_K v\, r\, dx, \quad r \in \mathbb{P}_{n-1}(K). \tag{1}$$

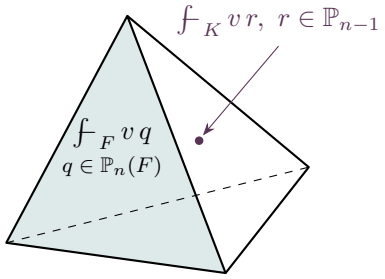
We use one functional per element of a basis of each test space, so

$$N_n = 4\binom{n+2}{2} + \binom{n+2}{3} = \frac{(n+1)(n+2)(n+12)}{6}. \tag{2}$$

Normalisation by area or volume does not affect rank or unisolvence; it makes the moments invariant under affine maps and vertex relabelling. Let $\mathcal{M}_n v \in \mathbb{R}^{N_n}$ collect (1). A local shape space $V_n(K)$ is sought with

$$\mathbb{P}_{n+1}(K) \subset V_n(K), \qquad \dim V_n(K) = N_n, \qquad \mathcal{M}_n|_{V_n(K)} \text{ bijective.} \tag{3}$$

The first condition gives order $n+1$ in the H^1 seminorm; the last is unisolvence. Figure 1 summarises the setting and the answer.



4 faces $\times\binom{n+2}{2} + \binom{n+2}{3}$ cell moments

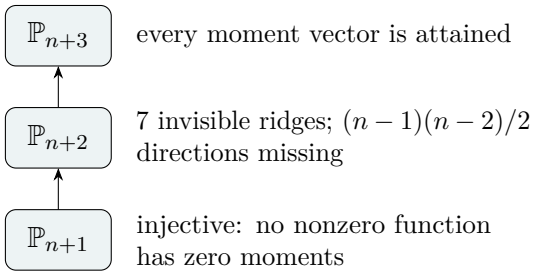


Figure 1: Left: the degrees of freedom on a tetrahedron. Right: what the moments see in successive complete polynomial spaces (Theorem 2.1). For $n = 3$ one direction is missing on $\mathbb{P}_5$, and it is supplied by the sextic Δ_4.

Theorem 2.1 (Moment rank and minimum degree)**.** *For every tetrahedron K and every $n \geq 1$:*

$$\ker(\mathcal{M}_n|_{\mathbb{P}_{n+1}(K)}) = \{0\}, \tag{4}$$

$$\dim\ker(\mathcal{M}_n|_{\mathbb{P}_{n+2}(K)}) = 7, \qquad \operatorname{rank}(\mathcal{M}_n|_{\mathbb{P}_{n+2}(K)}) = \binom{n+5}{3} - 7, \tag{5}$$

$$c_n := N_n - \operatorname{rank}(\mathcal{M}_n|_{\mathbb{P}_{n+2}(K)}) = \frac{(n-1)(n-2)}{2}, \tag{6}$$

$$\mathcal{M}_n|_{\mathbb{P}_{n+3}(K)} \text{ is surjective.} \tag{7}$$

Consequently the smallest r for which a space satisfying (3) exists inside $\mathbb{P}_r(K)$ is $r = n+2$ for $n = 1, 2$ and $r = n+3$ for $n \geq 3$. Such spaces can be chosen invariant under vertex relabelling and transported between tetrahedra by affine maps.

The kernel in (5) counts *functions* invisible to the moments; the number c_n in (6) counts *moment directions* that $\mathbb{P}_{n+2}$ cannot reach. They differ because $\dim\mathbb{P}_{n+2} \neq N_n$. The minimum-degree statement concerns single-cell polynomial spaces satisfying (3); it is not a statement about all fourth-order Stokes elements.

Table 2: Consequences of Theorem 2.1 [proof; each entry also checked by exact computation].

n	N_n	$\dim\mathbb{P}_{n+1}$	$\dim\mathbb{P}_{n+2}$	kernel	rank on $\mathbb{P}_{n+2}$	c_n
1	13	10	20	7	13	0
2	28	20	35	7	28	0
3	50	35	56	7	49	1
4	80	56	84	7	77	3
5	119	84	120	7	113	6
6	168	120	165	7	158	10

Example 2.2 (The smallest cases)**.** For $n = 1$ there are $N_1 = 13$ moments: three per face and one cell average. They are injective on the ten quadratics. On the twenty cubics, the kernel is spanned by seven cubics such as $U_0 = \lambda_0(5\lambda_0^2 - 5\lambda_0 + 1)$. This function vanishes on F_0. On a face F_j, $j \neq 0$, Lemma 4.4 below reduces its moments against 1, λ_0 and either other coordinate to multiples of

$$\int_0^1 t(5t^2 - 5t + 1)(1-t)\,dt, \quad \int_0^1 t^2(5t^2 - 5t + 1)(1-t)\,dt, \quad \int_0^1 t(5t^2 - 5t + 1)(1-t)^2\,dt,$$

all of which are zero; the cell average reduces to the last integral. Since $20 - 7 = 13 = N_1$, cubics reach every moment vector. For $n = 3$ the same count gives $56 - 7 = 49 < 50$: one direction is missing.

3 The fourth-order obstruction and its sextic repair

This section does not use the kernel theorem. Put

$$\Delta_4 = \prod_{0\le i<j\le 3} (\lambda_j - \lambda_i), \qquad \Delta_3^{(i)} = \prod_{\substack{j<k\\ j,k\neq i}} (\lambda_k - \lambda_j), \qquad b_i = \prod_{j\neq i} \lambda_j,$$

and define the signed cofactors

$$\varepsilon_i = (-1)^i \Delta_3^{(i)} = -\frac{\Delta_4}{\prod_{j\neq i}(\lambda_i - \lambda_j)}. \tag{8}$$

The second expression follows from separating the factors of Δ_4 that contain λ_i: those with $j < i$ give $\lambda_i - \lambda_j$ and the $3 - i$ factors with $k > i$ give $\lambda_k - \lambda_i$. The alternating face functional is

$$\mathcal{A}(v) = \sum_{i=0}^{3} \fint_{F_i} v\, \varepsilon_i \, ds. \tag{9}$$

For $n = 3$ each term is a face moment, so $\mathcal{A}$ is a linear combination of the entries of $\mathcal{M}_3 v$.

Lemma 3.1 (Sign rule)**.** *For a permutation σ of $\{0, 1, 2, 3\}$ write $(\sigma\lambda)_j = \lambda_{\sigma(j)}$ and $(v \circ \sigma)(\lambda) = v(\sigma\lambda)$. Then $\varepsilon_i(\sigma\lambda) = \operatorname{sgn}(\sigma)\, \varepsilon_{\sigma(i)}(\lambda)$ and $\mathcal{A}(v \circ \sigma) = \operatorname{sgn}(\sigma)\mathcal{A}(v)$.*

Proof. Since $\Delta_4(\sigma\lambda) = \operatorname{sgn}(\sigma)\Delta_4(\lambda)$ and $\prod_{j\neq i}(\lambda_{\sigma(i)} - \lambda_{\sigma(j)}) = \prod_{k\neq\sigma(i)}(\lambda_{\sigma(i)} - \lambda_k)$, the second form of (8) gives the first identity. The substitution $\mu = \sigma\lambda$ is an affine automorphism of K that preserves normalised measures and maps F_i onto $F_{\sigma^{-1}(i)}$. Using the first identity with σ^{-1},

$$\mathcal{A}(v \circ \sigma) = \sum_i \fint_{F_{\sigma^{-1}(i)}} v(\mu)\, \operatorname{sgn}(\sigma)\, \varepsilon_{\sigma^{-1}(i)}(\mu)\, ds = \operatorname{sgn}(\sigma)\mathcal{A}(v). \qquad \square$$

Proposition 3.2 (An explicit missing direction)**.** *$\mathcal{A}$ vanishes on $\mathbb{P}_5(K)$, and $\mathcal{A}(\Delta_4) = 1/34650$. Hence the 50 moments $\mathcal{M}_3$ are linearly dependent on $\mathbb{P}_5(K)$ and on every subspace of it.*

Proof. Represent $v \in \mathbb{P}_5$ by a homogeneous quintic in $\lambda_0, \dots, \lambda_3$, multiplying lower-degree parts by powers of $\sum_i \lambda_i = 1$. Its antisymmetrisation $v^{\mathrm{alt}} = \frac{1}{24}\sum_\sigma \operatorname{sgn}(\sigma) v \circ \sigma$ is an alternating homogeneous quintic. Setting $\lambda_i = \lambda_j$ shows that each $\lambda_j - \lambda_i$ divides it; these linear forms are pairwise coprime, so Δ_4, of degree six, divides it, and $v^{\mathrm{alt}} = 0$. By Lemma 3.1, $\mathcal{A}(v) = \mathcal{A}(v^{\mathrm{alt}}) = 0$.

On F_i, (8) with $\lambda_i = 0$ gives $\Delta_4|_{F_i} = -\varepsilon_i \prod_{j\neq i}(-\lambda_j) = \varepsilon_i b_i$, so

$$\mathcal{A}(\Delta_4) = \sum_{i=0}^{3} \fint_{F_i} b_i\, (\Delta_3^{(i)})^2\, ds > 0.$$

On a triangle with barycentrics a, b, c, a homogeneous polynomial h of degree k satisfies $\fint_F h = \frac{2}{(k+2)!}\int_{\mathbb{R}^3_+} h(x) e^{-x_1-x_2-x_3} dx$, because both sides equal $2\alpha!\beta!\gamma!/(k+2)!$ on monomials. With $h = abc\,\Delta_3^2$ ($k = 9$), the Andreief identity [13] for the weight $w(x) = xe^{-x}$ gives

$$\int_{\mathbb{R}^3_+} x_1x_2x_3\Delta_3(x)^2 e^{-\sum x_i} dx = 3!\det\big[(i+j+1)!\big]_{i,j=0}^{2} = 6\det\begin{pmatrix} 1 & 2 & 6 \\ 2 & 6 & 24 \\ 6 & 24 & 120 \end{pmatrix} = 144.$$

Thus each face contributes $2 \cdot 144/11! = 1/138600$, and four faces give $1/34650$. $\square$

Changing basis, using nonmonomial quintics, or choosing other quintic enrichments cannot help: $\mathcal{A}$ vanishes on all of $\mathbb{P}_5$. This answers the question of Appendix A of [3] in the negative. With Theorem 2.1, the rank of $\mathcal{M}_3$ on $\mathbb{P}_5$ is exactly 49, so

$$W_* := \mathbb{P}_5(K) \oplus \operatorname{span}\{\Delta_4\}, \qquad \dim W_* = 57, \tag{10}$$

carries all 50 moments; seven kernel directions remain to be removed (Section 5.5).

4 The seven-dimensional kernel

We first show that at most seven polynomials of degree $m = n + 2$ are invisible to the moments and then exhibit seven. Put $Z_m = \ker(\mathcal{M}_{m-2}|_{\mathbb{P}_m(K)})$, $m \geq 3$. Kernel dimensions are affine invariants, so we work on $T = \{x \in \mathbb{R}^3 : x_i \geq 0,\ x_1 + x_2 + x_3 \leq 1\}$ with $\lambda_i = x_i$ $(i = 1, 2, 3)$ and $\lambda_0 = 1 - x_1 - x_2 - x_3$.

4.1 What vanishing moments say about derivatives

Lemma 4.1 (Gradient orthogonality)**.** *If $p \in Z_m$, then $\int_T (\partial_i p) q\, dx = 0$ for all $q \in \mathbb{P}_{m-2}(T)$, $i = 1, 2, 3$. Moreover $Z_m \cap \mathbb{P}_{m-1}(T) = \{0\}$; hence $p \mapsto$ (homogeneous part of degree m) is injective on Z_m.*

Proof. $\int_T (\partial_i p) q = \sum_F \int_F p\, q\, \eta_{F,i}\, ds - \int_T p\, \partial_i q$, with face tests of degree $\leq m-2 = n$ and cell tests of degree $\leq m-3 = n-1$; both vanish. If also $p \in \mathbb{P}_{m-1}$, then $\int_T |\nabla p|^2 = \sum_F \int_F p\, \partial_{\boldsymbol{\eta}} p - \int_T p\, \Delta p = 0$ by the same degree count, so p is constant, and its face averages force $p = 0$ (compare [23, Thm. 3.1]). □

4.2 A classical simplex operator

Let $\mathcal{E} = \sum_i x_i \partial_i$, $\mathcal{D} = \sum_i x_i \partial_i^2 + \sum_i \partial_i$ and $\mathcal{L} = \mathcal{D} - \mathcal{E}^2 - 3\mathcal{E}$. Then $\mathcal{E}$ multiplies a homogeneous polynomial of degree r by r, $\mathcal{D}$ lowers degree by one, and

$$\mathcal{L} q = \operatorname{div}(C \nabla q), \qquad C(x) = \operatorname{diag}(x) - x x^{\mathsf{T}}, \tag{11}$$

which is the uniform-weight simplex Jacobi operator [1, Eq. (1.2)]. Indeed $\sum_i \partial_i (x_i \partial_i q) = \mathcal{D} q$ and $\sum_i \partial_i (x_i\, \mathcal{E} q) = 3\mathcal{E} q + \mathcal{E}^2 q$.

Lemma 4.2 (Orthogonal layers are eigenspaces)**.** *If $q \in \mathbb{P}_r(T)$, $r \geq 1$, is $L^2(T)$-orthogonal to $\mathbb{P}_{r-1}(T)$, then $\mathcal{L} q = -r(r+3) q$.*

Proof. The flux $C \nabla q \cdot \boldsymbol{\eta}$ vanishes on ∂T: on $x_i = 0$ the i-th component of $C \nabla q$ is zero, and on $\sum x_i = 1$ the sum of its components is $(1 - \sum x_i)\, \mathcal{E} q = 0$. So $\mathcal{L}$ is symmetric in $L^2(T)$. It maps $\mathbb{P}_r$ to itself, with leading part $-r(r+3)$ times the leading part of q, so $g = (\mathcal{L} + r(r+3)) q \in \mathbb{P}_{r-1}$ and $\int_T g^2 = \int_T q\, (\mathcal{L} + r(r+3)) g = 0$. □

4.3 Compatibility leaves seven coefficient patterns

Lemma 4.3 (Upper bound)**.** *For $m \geq 3$, $\dim Z_m \leq 7$.*

Proof. Write $p = H + H' + (\text{degree} \leq m-2)$ with H, H' homogeneous of degrees $m, m-1$. By Lemmas 4.1 and 4.2, $(\mathcal{L} + (m-1)(m+2)) \partial_i p = 0$. The terms of degree $m-2$ give

$$\mathcal{D}(\partial_i H) + 2m\, \partial_i H' = 0, \qquad i = 1, 2, 3, \tag{12}$$

since $(m-1)(m+2)-(m-2)(m+1)=2m$. Apply ∂_j to the i-th equation and ∂_i to the j-th and subtract: the H' terms cancel, and $[\partial_j, \mathcal{D}] = \partial_j^2$ gives

$$\partial_i\partial_j(\partial_j - \partial_i)H = 0, \qquad i \neq j. \tag{13}$$

Write $H = \sum_{|\alpha|=m} c_\alpha x^\alpha/\alpha!$. Then (13) says $c_{\beta+e_i+2e_j} = c_{\beta+2e_i+e_j}$ for $|\beta| = m-3$: one unit of exponent may move between two positive coordinates, keeping both positive, without changing the coefficient. Such moves connect all exponents with the same support, so there is at most one free coefficient per nonempty support $S \subset \{1,2,3\}$, seven in all. Lemma 4.1 transfers the bound to Z_m. □

The seven forms $(\sum_{i\in S} x_i)^m$, $\varnothing \neq S \subset \{1,2,3\}$, satisfy (13) and are independent: choose one monomial of degree m with support exactly T for each T; its coefficient in $(\sum_S x_i)^m$, divided by the multinomial coefficient, is $\mathbf{1}[T \subseteq S]$. Ordered by decreasing $|T|$ this matrix is unitriangular.

4.4 Seven explicit ridge polynomials

The integrals below all reduce to one variable by the following classical property of the uniform distribution on a simplex.

Lemma 4.4 (Integration along a barycentric sum)**.** *Let S be a d-simplex with barycentrics $\lambda_0, \dots, \lambda_d$, let A be a nonempty proper subset of the labels, $a = |A|$, $b = d+1-a$, $s = \sum_{i\in A}\lambda_i$. For a multi-index α and an integrable φ on $[0,1]$,*

$$⨍_S \varphi(s)\lambda^\alpha = c_\alpha \int_0^1 \varphi(s)\, s^{a-1+|\alpha_A|}(1-s)^{b-1+|\alpha_{A^c}|}\, ds,$$

with $c_\alpha > 0$ independent of φ.

Proof. Under the normalised measure, $(\lambda_0, \dots, \lambda_d)$ is Dirichlet$(1, \dots, 1)$ distributed. By the aggregation property of the Dirichlet law, s has density proportional to $s^{a-1}(1-s)^{b-1}$, and s, $(\lambda_i/s)_{i\in A}$ and $(\lambda_i/(1-s))_{i\notin A}$ are independent. Write $\lambda^\alpha = s^{|\alpha_A|}(1-s)^{|\alpha_{A^c}|}\prod_{i\in A}(\lambda_i/s)^{\alpha_i}\prod_{i\notin A}(\lambda_i/(1-s))^{\alpha_i}$ and factor the expectation. □

Applied to a face F_j (a 2-simplex), the exponent of s drops by one if $j \in A$ and that of $1-s$ drops by one if $j \notin A$. We use the Jacobi polynomials $J_r^{(\alpha,\beta)}$, for which $J_r^{(\alpha,\beta)}(2t-1)$ is orthogonal on $[0,1]$ to $\mathbb{P}_{r-1}$ with weight $(1-t)^\alpha t^\beta$ [12, §18.3], and $J_r = J_r^{(0,0)}$. A *ridge* is a polynomial that depends on a single affine combination of the coordinates. Define

$$U_i = \lambda_i\, J_{m-1}^{(1,1)}(2\lambda_i - 1)\ (i = 0, \dots, 3), \qquad V_{0j} = J_m(2(\lambda_0 + \lambda_j) - 1)\ (j = 1, 2, 3). \tag{14}$$

Lemma 4.5 (Kernel basis)**.** *For $m \geq 3$, the seven polynomials (14) form a basis of Z_m.*

Proof. Use monomial tests λ^α and Lemma 4.4. For U_i take $A = \{i\}$ $(a = 1, b = 3)$. On F_i, $U_i = 0$. On F_j, $j \neq i$, with $|\alpha| \leq m-2$, the moment is a multiple of $\int_0^1 t(1-t)J_{m-1}^{(1,1)}(2t-1)\, t^{\alpha_i}(1-t)^{|\alpha|-\alpha_i}dt = 0$, a test of degree $\leq m-2$. In the cell, $|\alpha| \leq m-3$ and the weight $(1-t)^2$ leave a test $t^{\alpha_i}(1-t)^{|\alpha|-\alpha_i+1}$ of degree $\leq m-2$. For V_{0j} take $A = \{0, j\}$ $(a = b = 2)$. Face weights are s or $1-s$ and face tests have degree $\leq m-2$; the cell weight is $s(1-s)$ and cell tests have degree $\leq m-3$. Either way J_m meets a polynomial of degree $\leq m-1$. Hence all seven lie in Z_m. Their leading parts are nonzero multiples of $x_1^m, x_2^m, x_3^m, (x_1+x_2+x_3)^m$ and $(x_1+x_2)^m, (x_1+x_3)^m, (x_2+x_3)^m$ (for V_{0j}, $\lambda_0 + \lambda_j = 1 - \sum_{k\neq j} x_k$ and the Legendre parity $J_m(-t) = (-1)^m J_m(t)$), which are independent. Lemma 4.3 completes the proof. □

Remark 4.6 (Low-order precedents). For $m = 3, 4$ the vertex ridges are proportional to $5\lambda_i^3 - 5\lambda_i^2 + \lambda_i$ and $14\lambda_i^4 - 21\lambda_i^3 + 9\lambda_i^2 - \lambda_i$, which appear in [9, Eqs. (4), (11)]. The argument above concerns the combined face and cell moments and gives completeness for every m.

5 Sharp degree, minimal enrichment and symmetric elements

5.1 Degree $n+3$ suffices

Lemma 5.1 (Surjectivity on $\mathbb{P}_{n+3}$)**.** *For every $n \geq 1$, $\mathcal{M}_n|_{\mathbb{P}_{n+3}(K)}$ is surjective.*

Proof. We show that the moment functionals are linearly independent on $\mathbb{P}_{n+3}$. Suppose $\sum_i \int_{F_i} v\, q_i\, ds + \int_K v\, r\, dx = 0$ for all $v \in \mathbb{P}_{n+3}$, with $q_i \in \mathbb{P}_n(F_i)$, $r \in \mathbb{P}_{n-1}(K)$. The test $v = \lambda_0\lambda_1\lambda_2\lambda_3 r$ (degree $\leq n+3$) vanishes on ∂K and gives $\int_K \lambda_0 \cdots \lambda_3 r^2 = 0$, so $r = 0$. For a face F_i, extend q_i to $\tilde{q}_i \in \mathbb{P}_n(K)$; the test $v = b_i\tilde{q}_i$ vanishes on the other faces and gives $\int_{F_i} b_i q_i^2 = 0$, so $q_i = 0$. □

Proof of Theorem 2.1. Lemma 4.1 gives (4), Lemma 4.5 and rank–nullity give (5), and $N_n - \binom{n+5}{3} + 7 = (n-1)(n-2)/2$ gives (6); Lemma 5.1 is (7). Degree $n+1$ is too small since $N_n - \dim \mathbb{P}_{n+1} = n(n+2) > 0$. If $\mathcal{M}_n$ is surjective on $\mathbb{P}_r$, extend a basis of $\mathbb{P}_{n+1}$ (on which $\mathcal{M}_n$ is injective) by a complement of $\ker \mathcal{M}_n|_{\mathbb{P}_r}$ chosen to contain $\mathbb{P}_{n+1}$; this gives (3) in $\mathbb{P}_r$. Surjectivity holds on $\mathbb{P}_{n+2}$ exactly when $c_n = 0$, that is $n \leq 2$, and always on $\mathbb{P}_{n+3}$. Invariance is Proposition 5.3. □

5.2 How many top-degree functions are needed

Corollary 5.2 (Minimal enrichment count)**.** *Let $n \geq 1$ and let V be any polynomial space satisfying (3). Then $\dim((V + \mathbb{P}_{n+2})/\mathbb{P}_{n+2}) \geq c_n$: every such space contains at least $(n-1)(n-2)/2$ independent directions outside $\mathbb{P}_{n+2}$. This number is attained by vertex-invariant spaces inside $\mathbb{P}_{n+3}$ (Theorem 5.4).*

Proof. Choose $V' \subset V$ complementary to $V \cap \mathbb{P}_{n+2}$. Then $\mathbb{R}^{N_n} = \mathcal{M}_n(V) \subset \mathcal{M}_n(\mathbb{P}_{n+2}) + \mathcal{M}_n(V')$, so $N_n \leq (N_n - c_n) + \dim V'$, and $\dim V' = \dim((V + \mathbb{P}_{n+2})/\mathbb{P}_{n+2})$. □

5.3 A numbering-free choice of complement

Regard barycentric polynomials as functions on $\sum_i \lambda_i = 1$, and put

$$\partial_{ij} = \frac{\partial}{\partial \lambda_i} - \frac{\partial}{\partial \lambda_j}, \qquad a_B(u, v) = \fint_K \sum_{0 \leq i < j \leq 3} \partial_{ij} u\, \partial_{ij} v\, dx. \tag{15}$$

Tangent derivatives are well defined on the hyperplane and span its tangent space, so a_B is positive semidefinite with kernel the constants. It is invariant under vertex permutations and affine maps. On a general tetrahedron it is a constant positive-definite metric applied to the physical gradient, not the Euclidean Dirichlet form.

Proposition 5.3 (Canonical complement)**.** *Let $W \supset \mathbb{P}_{n+1}(K)$ be a vertex-invariant polynomial space with $\mathcal{M}_n|_W$ surjective, $Z = \ker(\mathcal{M}_n|_W)$, and $V = \{v \in W : a_B(v, z) = 0\ \forall z \in Z\}$. Then $W = V \oplus Z$, $\dim V = N_n$, $\mathbb{P}_{n+1} \subset V$, $\mathcal{M}_n|_V$ is bijective, and V is invariant under vertex relabelling and affine pullback.*

Proof. No nonzero constant lies in Z, so a_B is positive definite on Z and $W = V \oplus Z$; bijectivity and the dimension follow. For $p \in \mathbb{P}_{n+1}$, $a_B(p, z) = \fint_K \nabla z \cdot G \nabla p$ with a constant symmetric matrix G; integrating by parts places on z face tests $G\nabla p \cdot \boldsymbol{\eta} \in \mathbb{P}_n(F)$ and the cell test $\operatorname{div}(G\nabla p) \in \mathbb{P}_{n-1}$, so $a_B(p, z) = 0$. Invariance holds because Z and a_B are invariant. □

With $W = \mathbb{P}_{n+2}$ for $n = 1, 2$ this gives numbering-free elements of minimal degree, answering the concern of Remark 3.6 of [3]; they differ from the non-symmetric spaces listed there. For $n \geq 3$, $W = \mathbb{P}_{n+3}$ works but uses many top-degree directions. The next subsection removes this waste.

5.4 Lean symmetric elements for every order

The symmetric group $\mathfrak{S}_4$ acts on polynomials by relabelling vertices and on moment vectors by permuting faces and transforming the tests; $\mathcal{M}_n$ is equivariant. Hence the space of missing directions $C_n = \mathbb{R}^{N_n}/\mathcal{M}_n(\mathbb{P}_{n+2})$ is an $\mathfrak{S}_4$-module of dimension c_n.

Theorem 5.4 (Lean symmetric elements)**.** *For every $n \geq 3$ there is a vertex-invariant space $E_n \subset \mathbb{P}_{n+3}(K)$ with $\dim E_n = c_n$ and $E_n \cap \mathbb{P}_{n+2} = \{0\}$ such that $W_n = \mathbb{P}_{n+2} \oplus E_n$ satisfies: $\mathcal{M}_n|_{W_n}$ is surjective and $\ker(\mathcal{M}_n|_{W_n}) = Z_{n+2}$. The canonical complement $V_n \subset W_n$ of Proposition 5.3 satisfies (3), is invariant, and has exactly c_n independent directions outside $\mathbb{P}_{n+2}$, the minimum of Corollary 5.2.*

Proof. Let Q be the $L^2(K)$-orthogonal complement of $\mathbb{P}_{n+2}$ in $\mathbb{P}_{n+3}$; it is invariant because normalised L^2 is. The composite $Q \to \mathbb{R}^{N_n} \to C_n$ is equivariant, and surjective by Lemma 5.1. By Maschke's theorem [22] its kernel has an invariant complement $E_n \subset Q$, mapped isomorphically onto C_n. Then $\mathcal{M}_n(W_n) = \mathcal{M}_n(\mathbb{P}_{n+2}) + \mathcal{M}_n(E_n) = \mathbb{R}^{N_n}$, and $\dim \ker = \dim \mathbb{P}_{n+2} + c_n - N_n = 7$; the kernel contains Z_{n+2}, so equals it. Since $Z_{n+2} \subset \mathbb{P}_{n+2}$, $V_n + \mathbb{P}_{n+2} = W_n$. □

The theorem is existential. The next result identifies the module and suggests an explicit choice.

Proposition 5.5 (The missing directions as a representation)**.** *For $n \geq 3$, $C_n \cong \operatorname{sgn} \otimes \operatorname{Sym}^{n-3}(\mathrm{std})$, where* std *is the three-dimensional standard representation of $\mathfrak{S}_4$. The space $\Delta_4 \mathbb{P}_{n-3}(K)$ modulo $\mathbb{P}_{n+2}$ is isomorphic to the same module.*

Proof. By Theorem 2.1, C_n equals, as a virtual module, $\mathbb{R}^{N_n} - \mathbb{P}_{n+2} + Z_{n+2}$. The characters of the face part (induced from the vertex stabiliser), of $\mathbb{P}_{n-1}$ and $\mathbb{P}_{n+2}$ (Molien series of the permutation action on $\lambda_0, \dots, \lambda_3$), and of Z_{n+2} (vertex ridges permute like vertices; pair ridges permute like the three bipartitions, with the sign $(-1)^{n+2}$ when the two parts are exchanged) have rational generating functions in n. Explicitly, for $g \in \mathfrak{S}_4$ let $f_g(t) = \prod_{\text{cycles } \gamma} (1 - t^{|\gamma|})^{-1}$ (cycles of g on the four vertices), let ϕ_g be the number of fixed vertices, let f'_g be the same product over the cycles of g on the three vertices other than one fixed vertex, and let $\beta_g^{\pm}$ be the numbers of bipartitions $\{A, A^c\}$ into pairs fixed by g with $gA = A$ $(+)$ or $gA = A^c$ $(-)$. Then

$$\sum_{n \geq 1} \chi_{C_n}(g)\, t^n = \phi_g (f'_g(t) - 1) + t f_g(t) - \frac{f_g(t) - [f_g]_{\leq 2}(t)}{t^2} + \frac{(\phi_g + \beta_g^+)\, t}{1 - t} - \frac{\beta_g^-\, t}{1 + t},$$

where $[f]_{\leq 2}$ is the Taylor polynomial of degree two; the five terms are the face part, $\mathbb{P}_{n-1}$, $-\mathbb{P}_{n+2}$, the vertex ridges plus the bipartitions kept, and those exchanged (sign $(-1)^{n+2}$). The character of $\operatorname{sgn} \otimes \operatorname{Sym}^{n-3}(\mathrm{std})$ has generating function $\operatorname{sgn}(g)\, t^3(1-t) f_g(t)$, because the permutation representation is triv $\oplus$ std. With $(\phi_g, \beta_g^+, \beta_g^-)$ and $\operatorname{sgn}(g) t^3 (1-t) f_g(t)$ equal to

$$\begin{array}{ll} e: & (4,3,0),\ \ t^3/(1-t)^3, \\ (01): & (2,1,0),\ \ -t^3/((1-t)^2(1+t)), \\ (01)(23): & (0,1,2),\ \ t^3/((1-t)(1+t)^2), \\ (012): & (1,0,0),\ \ t^3/(1-t^3), \\ (0123): & (0,0,1),\ \ -t^3/((1+t)(1+t^2)), \end{array}$$

the two sides agree as rational functions on each class [exact symbolic computation; script `verify_lean_element.py`]. For example, on the identity both sides reduce to $\sum_n \binom{n-1}{2} t^n$. For the second statement, $\mathbb{P}_r(K)$ is identified with homogeneous polynomials of degree r in $\lambda_0, \dots, \lambda_3$, and $\mathbb{P}_{r-1}$ with multiples of $e = \sum_i \lambda_i$. Since e does not divide Δ_4, $\Delta_4 f$ is a multiple of e only when f is, so $\Delta_4 \mathbb{P}_{n-3} / (\Delta_4 \mathbb{P}_{n-3} \cap \mathbb{P}_{n+2}) \cong \mathrm{sgn} \otimes \mathrm{Sym}^{n-3}(V/\langle e \rangle)$ with $V/\langle e \rangle \cong \mathrm{std}$. Its dimension is $\binom{n}{3} - \binom{n-1}{3} = c_n$. □

Conjecture 5.6 (Explicit lean enrichment)**.** *For every $n \geq 3$, $E_n = \Delta_4 \mathbb{P}_{n-3}(K)$ (modulo $\mathbb{P}_{n+2}$) can be used in Theorem 5.4, that is, $\mathcal{M}_n$ is surjective on $\mathbb{P}_{n+2} + \Delta_4 \mathbb{P}_{n-3}$.*

This holds for $n = 3$ by Proposition 3.2 [proof] and for $n = 3, \dots, 8$ by exact computation: the spaces have dimensions $57, 87, 126, 175, 235, 307$ and moment rank N_n [exact computation]. For $n = 3, 4$ the module is irreducible, so by Schur's lemma a single nonzero value suffices.

Example 5.7 (An explicit lean element for $n = 4$)**.** Let Π_6 be the a_B-orthogonal projection of $\mathbb{P}_6$ onto the complement of Z_6. The space

$$V_4^\flat = \Pi_6(\mathbb{P}_6) + \mathrm{span}\{\lambda_0 \Delta_4, \lambda_1 \Delta_4, \lambda_2 \Delta_4, \lambda_3 \Delta_4\}$$

has dimension $77 + 4 - 1 = 80$ (the four products sum to $\Delta_4 \in \mathbb{P}_6$), contains $\mathbb{P}_5$, is unisolvent for the 80 moments $\mathcal{M}_4$ and is vertex-invariant; it uses three degree-7 directions, the minimum $c_4 = 3$ [exact computation]. The products $\lambda_i \Delta_4$ are a_B-orthogonal to Z_6 by Schur's lemma, since they span $\mathrm{sgn} \oplus (\mathrm{sgn} \otimes \mathrm{std})$ while Z_6 contains only trivial, standard and two-dimensional components; this is also checked exactly. Similarly, for $n = 5$ the space $\Pi_7(\mathbb{P}_7) + \Delta_4 \mathbb{P}_2$ has dimension 119, contains $\mathbb{P}_6$, is unisolvent for $\mathcal{M}_5$ and is invariant [exact computation].

5.5 The 50-dimensional fourth-order element

For $n = 3$, $m = 5$, the kernel of $\mathcal{M}_3$ on W_* in (10) is Z_5 (an element with a nonzero Δ_4 component would have $\mathcal{A} \neq 0$). Moreover $a_B(\Delta_4, p) = 0$ for $p \in \mathbb{P}_5$: by invariance the pairing equals the pairing of Δ_4 with the antisymmetrisation of p, which is zero. Hence:

Corollary 5.8 (One-sextic fourth-order element)**.** *The space*

$$V_3^*(K) = \{p \in \mathbb{P}_5(K) : a_B(p, z) = 0 \ \forall z \in Z_5\} \oplus \mathrm{span}\{\Delta_4\} \tag{16}$$

has dimension 50, contains $\mathbb{P}_4(K)$, is unisolvent for $\mathcal{M}_3$, and is invariant under vertex relabelling and affine maps. Every polynomial space satisfying (3) for $n = 3$ has degree at least six and at least one direction outside $\mathbb{P}_5$, so V_3^* is minimal in both senses.

This is the minimal-rank repair: it uses the least possible number of sextic directions. We do not classify all fourth-order spaces.

Gram matrix. Let $\kappa = (U_0, U_1, U_2, U_3, V_{01}, V_{02}, V_{03})$ with $m = 5$, and put $f(t) = t J_4^{(1,1)}(2t-1)$, $g(s) = J_5(2s-1)$. From $\partial_{ab} U_i = f'(\lambda_i)(\mathbf{1}[a = i] - \mathbf{1}[b = i])$ and $\partial_{ab} V_A = g'(s_A)(\mathbf{1}[a \in A] - \mathbf{1}[b \in A])$:

- $a_B(U_i, U_i) = 3 \fint f'(\lambda_i)^2$ and $a_B(U_i, U_j) = -\fint f'(\lambda_i) f'(\lambda_j)$ $(i \neq j)$, one- and two-variable integrals against the densities $3(1-t)^2$ and $6(1-t-u)$ (Lemma 4.4), equal to $225/11$ and $-5/11$;
- $a_B(V_A, V_A) = 4 \fint g'(s)^2 = 4 \int_0^1 6s(1-s) g'(s)^2 ds = 720/11$, by $\int_{-1}^1 (1-x^2) J_5'(x)^2 dx = 2 \cdot 30/11$; distinct pair ridges share two crossing edges with opposite signs, so are a_B-orthogonal;
- $a_B(U_i, V_A) = \pm 2 \fint f'(\lambda_i) g'(s_A) = \pm 60/11$, with $+$ if $i \in A$.

The values 225/11, −5/11 and 60/11 are evaluated in exact arithmetic. Thus

$$\Gamma = \frac{1}{11}\begin{pmatrix} 230I_4 - 5J_4 & 60S \\ 60S^{\mathsf{T}} & 720I_3 \end{pmatrix}, \qquad \Gamma^{-1} = 11\begin{pmatrix} I_4/210 & -S/2520 \\ -S^{\mathsf{T}}/2520 & 23I_3/15120 \end{pmatrix}, \\ S = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}, \tag{17}$$

where J_4 is the all-ones matrix; the inverse follows from $S^{\mathsf{T}}S = 4I_3$, $SS^{\mathsf{T}} = 4I_4 - J_4$, $J_4S = 0$.

Projection. For $q \in \mathbb{P}_5$ let $\Pi q = q - \kappa^{\mathsf{T}}\Gamma^{-1}(a_B(\kappa_1, q), \dots, a_B(\kappa_7, q))^{\mathsf{T}}$. This is the identity on $\mathbb{P}_4$ and maps $\mathbb{P}_5$ onto the first summand of (16).

Proposition 5.9 (A 14-monomial basis)**.** *Let Q_{14} be the quintics $x_1^a x_2^b x_3^c$ with (a, b, c) in {(0,1,4), (0,2,3), (0,3,2), (1,0,4), (1,1,3), (1,2,2), (1,3,1), (1,4,0), (2,0,3), (2,1,2), (2,2,1), (2,3,0), (3,0,2), (3,2,0)}. A basis of $\mathbb{P}_4$, the fourteen Πq ($q \in Q_{14}$) and Δ_4 form a basis of $V_3^*(K)$.*

Proof. The seven excluded quintic exponents $(5,0,0)$, $(0,5,0)$, $(0,0,5)$, $(4,1,0)$, $(4,0,1)$, $(0,4,1)$, $(3,1,1)$ have one exponent for each nonempty support $S \subset \{1,2,3\}$. Modulo $\mathbb{P}_4$ and span Q_{14}, the leading form $(\sum_{i\in S} x_i)^5$ of the ridge with support S has coefficients on these seven monomials forming a lower triangular matrix with diagonal $(1,1,1,5,5,5,20)$ (the multinomial coefficients). Hence $\mathbb{P}_4 + \operatorname{span} Q_{14} + Z_5 = \mathbb{P}_5$, so $\Pi(\mathbb{P}_4 + \operatorname{span} Q_{14})$ is the 49-dimensional first summand. □

An exact certificate confirms the full 50×50 moment matrix of this basis (determinant $-1/(2^{174}3^{112}5^{65}7^{46}\,11)$). The determinant is tiny because of scaling, not because of near-dependence; see Section 9 for a well-conditioned basis.

6 Triangles and higher-dimensional simplices

On a d-simplex K_d, let $\mathcal{M}_{d,n}$ consist of all facet moments against $\mathbb{P}_n$ and cell moments against $\mathbb{P}_{n-1}$, $n \geq 1$, $m = n + 2$. Then $N_{d,n} = (d+1)\binom{n+d-1}{d-1} + \binom{n+d-1}{d}$. Put $k(d,m) = \sum_{r=1}^{\min(d,m)} \binom{d}{r}$.

Theorem 6.1 (Kernel on a d-simplex)**.** *For $d \geq 2$, $\mathcal{M}_{d,n}$ is injective on $\mathbb{P}_{m-1}(K_d)$, $\dim\ker(\mathcal{M}_{d,n}|_{\mathbb{P}_m}) = k(d,m)$, and it is surjective on $\mathbb{P}_{n+d}$.*

Proof. Lemma 4.1 holds verbatim. With $\mathcal{L}_d = \mathcal{D} - \mathcal{E}^2 - d\mathcal{E} = \operatorname{div}(C\nabla\cdot)$, orthogonal layers satisfy $\mathcal{L}_d q = -r(r+d)q$, and the degree-$(m-2)$ terms give $\mathcal{D}\partial_i H + (2m+d-3)\partial_i H' = 0$, hence (13). A degree-$m$ monomial has at most $\min(d,m)$ variables, so there are at most $k(d,m)$ free coefficients. For the lower bound, take a nonempty proper label set A, $a = |A|$, $b = d+1-a$, $s = \sum_A \lambda_i$, and define $R_A = J_m^{(b-2,a-2)}(2s-1)$ if $a, b \geq 2$, $R_A = s\,J_{m-1}^{(d-2,1)}(2s-1)$ if $a = 1$ (and the complementary rule if $b = 1$). By Lemma 4.4, facet and cell weights are the Jacobi weight times s, $1-s$ or $s(1-s)$, and the test degrees are low enough, as in Lemma 4.5. For $S \subset \{1,\dots,d\}$, $1 \leq |S| \leq \min(d,m)$, take A to be S (or the complement $\{0\}$ when $S = \{1,\dots,d\}$); the leading part of R_A is a nonzero multiple of $(\sum_S x_i)^m$, and the unitriangular support argument gives independence. Surjectivity on $\mathbb{P}_{n+d}$ follows as in Lemma 5.1 with the cell bubble (degree $d+1$) and facet bubbles (degree d). □

Proposition 6.2 (Missing directions in every dimension)**.** $c(d,n) := N_{d,n} - \operatorname{rank}(\mathcal{M}_{d,n}|_{\mathbb{P}_{n+2}})$ *equals*

$$c(d,n) = \sum_{r=3}^{\min(d,n)} \binom{d}{r}\Big[(r-2)\binom{n}{r-1} - \binom{n}{r-2} + 1\Big].$$

Every term is positive. Hence $c(d,n)=0$ if $d=2$ or $n\le 2$, and $c(d,n)>0$ if $d\ge 3$ and $n\ge 3$. For example $c(3,n)=(n-1)(n-2)/2$, $c(d,3)=\binom{d}{3}$, and $c(4,n)=(n-2)(n+5)(2n-3)/6$ for $n\ge 2$ ($c(4,1)=0$).

Proof. Write $h_r=\binom{r+d-1}{d-1}$ for the number of degree-r monomials in d variables. From $\binom{n+1+d}{d}=\binom{n+d-1}{d}+h_n+h_{n+1}$,

$$c(d,n)=d\,h_n-h_{n+1}-h_{n+2}+k(d,n+2).$$

Counting monomials by support size gives $h_r=\sum_{t\ge 1}\binom{d}{t}\binom{r-1}{t-1}$ $(r\ge 1)$. Using $d\binom{d}{t}=t\binom{d}{t}+(t+1)\binom{d}{t+1}$ and Pascal's rule, $d\,h_n=\sum_t\binom{d}{t}\,t\binom{n}{t-1}$. Collecting coefficients of $\binom{d}{t}$ and using $\binom{n+1}{t-1}=\binom{n}{t-1}+\binom{n}{t-2}$ gives the bracket $(t-2)\binom{n}{t-1}-\binom{n}{t-2}+\mathbf{1}[t\le n+2]$, which is zero for $t=1,2,n+1,n+2$ and for $t>n+2$. For $3\le t\le n$, $(t-2)\binom{n}{t-1}=\binom{n}{t-2}\frac{(t-2)(n-t+2)}{t-1}\ge\binom{n}{t-2}$ because $(t-2)(n-t+2)\ge 2(t-2)\ge t-1$, so the bracket is at least one. □

Corollary 6.3 (Triangles)**.** *For edge moments against $\mathbb{P}_n$ and cell moments against $\mathbb{P}_{n-1}$ on a triangle, the kernel on $\mathbb{P}_{n+2}$ is three-dimensional and the moments are surjective on $\mathbb{P}_{n+2}$ for every $n\ge 1$. A unisolvent space containing $\mathbb{P}_{n+1}$ is obtained by adding n functions of degree $n+2$.*

This is why triangular constructions in the spirit of [14, 18, 4] can stay one degree above the reproduced polynomials at every order, while on tetrahedra the extra degree becomes necessary at fourth order. In dimension $d\ge 3$ Theorem 6.1 and Proposition 6.2 give, for $n\ge 3$, a minimum degree between $n+3$ and $n+d$.

Conjecture 6.4 (Sharp degree in every dimension)**.** *For every $d\ge 3$ and $n\ge 1$, $\mathcal{M}_{d,n}$ is surjective on $\mathbb{P}_{n+3}(K_d)$. Hence the minimum degree is $n+2$ for $n\le 2$ and $n+3$ for $n\ge 3$.*

Table 3: Evidence for Conjecture 6.4 [exact computation]. Full row rank modulo the prime $2^{61}-1$ implies full rank over $\mathbb{Q}$, so each "yes" is a certificate of surjectivity.

d	n checked	$c(d,n)$ on $\mathbb{P}_{n+2}$	surjective on $\mathbb{P}_{n+3}$
2	1–6	0	yes
3	1–6	0,0,1,3,6,10	yes
4	1–5	0,0,4,15,35	yes
5	1–4	0,0,10,45	yes
6	1–3	0,0,20	yes

The ranks on $\mathbb{P}_{n+2}$ in Table 3 agree with Theorem 6.1. The bubble proof of Lemma 5.1 uses that cell and face bubbles have degrees 4 and 3, which is special to $d=3$; a proof of Conjecture 6.4 will need a different argument.

7 Stokes consequences

These results apply the established nonconforming framework of [11, 3]; we state them to fix hypotheses. Let $\Omega\subset\mathbb{R}^3$ be a bounded connected Lipschitz polyhedron, and $\mathcal{T}_h$ a conforming shape-regular family of affine tetrahedral meshes. Transport a reference space V_n from Theorem 2.1 (for example V_3^* or a lean space) to each cell. Let X_h consist of piecewise functions in $V_n(K)$ whose jumps on interior faces, and traces on boundary faces, are orthogonal to $\mathbb{P}_n(F)$; put $\mathbf{X}_h=X_h^3$ and $Q_h=\{q\in L_0^2(\Omega):q|_K\in\mathbb{P}_n(K)\}$. The broken seminorm $|v|_{1,h}^2=\sum_K\|\nabla v\|_{0,K}^2$ is a norm on X_h (zero gradients give piecewise constants, which shared face averages and the boundary condition force to vanish).

Lemma 7.1 (Fortin operator)**.** *Let I_h match the moments (1) on every cell. There is C_I, independent of h, with $|I_h v|_{1,h} \le C_I |v|_{1,\Omega}$ for $v \in H^1_0(\Omega)$, and $\sum_K \int_K \operatorname{div}(\mathbf{v} - I_h\mathbf{v}) q_h = 0$ for $\mathbf{v} \in H^1_0(\Omega)^3$, $q_h \in Q_h$.*

Proof. On the reference cell the moments are H^1-continuous (trace theorem) and the inverse into the polynomial space is bounded; constants are reproduced, so a Poincaré argument and affine scaling give the stability bound. Integration by parts writes $\int_K \operatorname{div}(\mathbf{v} - I_K\mathbf{v}) q_h$ as face moments against $q_h \boldsymbol{\eta}_K$ (degree $\le n$) and cell moments against ∇q_h (degree $\le n-1$), all zero. □

Corollary 7.2 (Discrete inf-sup)**.** *If $\beta_\Omega > 0$ is the continuous inf-sup constant, then* $\inf_{q_h} \sup_{\mathbf{v}_h} \frac{\sum_K \int_K q_h \operatorname{div} \mathbf{v}_h}{|\mathbf{v}_h|_{1,h} \|q_h\|_0} \ge \beta_\Omega / C_I$.

For viscosity $\nu > 0$ and $\mathbf{f} \in L^2(\Omega)^3$, the discrete problem uses $a_h(\mathbf{u}, \mathbf{v}) = \nu \sum_K \int_K \nabla\mathbf{u} : \nabla\mathbf{v}$ and $b_h(\mathbf{v}, q) = -\sum_K \int_K q \operatorname{div} \mathbf{v}$, with exact integration. This is the vector-Laplacian form; a symmetric-gradient form would need a discrete Korn inequality, which we do not study.

Proposition 7.3 (Energy-order convergence)**.** *If $\mathbf{u} \in H^{n+2}(\Omega)^3 \cap H^1_0$ and $p \in H^{n+1}(\Omega) \cap L^2_0$, then $|\mathbf{u} - \mathbf{u}_h|_{1,h} + \|p - p_h\|_0 \le C h^{n+1} (\|\mathbf{u}\|_{n+2} + \|p\|_{n+1})$, with C depending on n, Ω, ν and shape regularity. For V_3^* this is fourth order.*

Proof. Interpolation reproduces $\mathbb{P}_{n+1}$, which gives the approximation term. The consistency error consists of face terms $\int_F (\sigma\boldsymbol{\eta}_F) \cdot [\mathbf{v}_h]$ with $\sigma = \nu\nabla\mathbf{u} - pI$; since jumps are orthogonal to $\mathbb{P}_n(F)$, subtracting a face projection of $\sigma\boldsymbol{\eta}_F$ and using trace and Poincaré inequalities bounds it by $C h^{n+1} (\nu\|\mathbf{u}\|_{n+2} + \|p\|_{n+1}) |\mathbf{v}_h|_{1,h}$. The nonconforming saddle-point Strang estimate [6, Ch. 5] combines this with coercivity and the inf-sup bound. □

No claim is made about pressure robustness [17], curved cells, or p-uniform constants.

8 A numerical Stokes test

All results in this section are [numerical]. We used V_3^* on the unit cube, split into N^3 subcubes of six Kuhn tetrahedra each, with $\nu = 1$, the exact solution $\mathbf{u} = \operatorname{curl}(\phi, \phi, \phi)$, $\phi = \sin^2(\pi x)\sin^2(\pi y)\sin^2(\pi z)$, and $p = \cos(\pi x)\cos(\pi y)\cos(\pi z)$. The mean-zero pressure condition was imposed by a Lagrange multiplier. Polynomial products were integrated exactly: stiffness, divergence and mass terms with a collapsed Gauss–Jacobi (Stroud) rule with 8 points per direction (512 points, exact to degree 15), and face moments with the corresponding triangle rule with 5 points per direction (exact to degree 9). The non-polynomial load $(\mathbf{f}, \mathbf{v}_h)$ and all error norms were computed with the same fixed 512-point degree-15 rule on every cell, without adaptivity or a tolerance; we did not separately quantify this quadrature error. The system was solved by a sparse direct solver.

Table 4: Errors and observed rates for V_3^* with discontinuous $\mathbb{P}_3$ pressure [numerical]. h is the largest edge length.

N	h	velocity DOFs	$\lvert\mathbf{u} - \mathbf{u}_h\rvert_{1,h}$	rate	$\lVert\mathbf{u} - \mathbf{u}_h\rVert_0$	rate	$\lVert p - p_h\rVert_0$ (rate)
1	1.732	360	4.638	–	0.2791	–	0.9318 (–)
2	0.866	3600	0.6501	2.84	0.02457	3.51	0.1414 (2.72)
3	0.5774	12960	0.1239	4.09	0.003009	5.18	0.02268 (4.51)
4	0.4330	31680	0.04119	3.83	7.508e-4	4.83	0.007201 (3.99)
5	0.3464	63000	0.01727	3.90	2.520e-4	4.89	0.002924 (4.04)

The coarsest meshes are pre-asymptotic. The computed rates are consistent with, but do not by themselves prove, the fourth-order estimate. From $N = 3$ on, the broken H^1 velocity error and the pressure error decrease at rates close to four, as Proposition 7.3 predicts, and the L^2 velocity error at a rate approaching five. The discrete inf-sup constant, computed as the square root of the smallest positive generalised eigenvalue of $BA^{-1}B^{\mathsf{T}}q = \beta^2 Mq$ on mean-zero pressures, is 0.4017, 0.3817, 0.3743, 0.3702 for $N = 1, \dots, 4$. These values are consistent with the mesh-independent bound of the Fortin argument (Lemma 7.1); four meshes cannot establish mesh independence. On a single element shrinking by halves, moment interpolation of a smooth function showed rates 4.94 in L^2 and 3.97 in the H^1 seminorm, and a random quartic was reproduced to relative error below 10^{-12}.

9 Implementation remarks

Consistent face bases. Neighbouring cells must use the same test functions on a shared face. We write the face tests in the face's three barycentric coordinates ordered by global vertex index; since V_3^* and the lean spaces are invariant under relabelling, the local space does not depend on the cell's vertex order, and no further transformation of degrees of freedom is needed (compare the general discussion in [21]).

Well-conditioned basis. Exact determinants are a poor guide to conditioning. On the reference tetrahedron the 50×50 moment matrix has 2-norm condition number about 6.6×10^7 for the monomial-type basis of Proposition 5.9 with monomial tests, and about 7.6 for an L^2-orthonormal basis of V_3^* with orthonormal face and cell tests [numerical]. We therefore recommend computing an orthonormal basis of V_3^* once, in barycentric terms, and the nodal (degree-of-freedom dual) basis from it; because the moments are normalised, the same nodal basis serves every affine cell. The reference stiffness matrix in that nodal basis has condition number about 7.2×10^2 on the nonconstant subspace.

Quadrature. Stiffness integrals need degree 8 for the quintic part and degree 10 for products involving Δ_4; face moments need degree 9. Since only one basis function is sextic, the higher-degree rule can be restricted to its row and column.

Counts. Per velocity component the element has 10 degrees of freedom on each face and 10 in each cell; the cell ones can be condensed statically. On Kuhn meshes this is about 30 per cell per component (about 20 after condensation), with 20 pressure unknowns per cell. Conforming $\mathbb{P}_4$ Lagrange velocities need fewer unknowns but are not inf-sup stable with discontinuous $\mathbb{P}_3$ pressures on general meshes; split-mesh conforming elements [20, 15] and virtual elements [8] are the natural alternatives to compare in a dedicated study, which we have not done.

10 Conclusion and open problems

The face-and-cell moments that make nonconforming Stokes elements inf-sup stable see all of $\mathbb{P}_{n+1}$, miss exactly seven polynomials of degree $n + 2$, and see everything in degree $n + 3$. On tetrahedra this forces degree $n + 3$ from fourth order on, explains why the quintic search of [3] could not succeed, and yields numbering-free elements with the fewest possible top-degree functions. Open problems: a proof of Conjectures 5.6 and 6.4; other moment patterns, such as facet tests one degree below the shape functions as in [7]; discrete Korn inequalities and pressure-robust reconstructions for this family; and p-robust constants.

Statements

Use of AI tools. Generative AI tools were used in developing the arguments, the verification code and the text. All mathematical statements were checked by the author, and the proofs, exact computations and numerical results are reported with the evidence labels used throughout. Responsibility for the content rests with the author.

Data and code availability. The exact verifiers, the numerical Stokes code and their deterministic outputs are available in the accompanying archive, doi:10.5281/zenodo.23565002, with source at `https://github.com/ipitchford/tetrahedral-moment-stokes-elements`. A single script regenerates every certificate byte for byte and checks it against published SHA-256 sums.

Competing interests. None.

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